The Nonabelian Quantum Fourier Transform

The Fourier Transform: Part 2

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January 31, 2025

Table of Contents

Last week (`https://arxiv.org/pdf/1301.7007`)
(*Pretending to factor large numbers on a quantum computer*)

- Mathematical basis of the DFT
- The Quantum Fourier Transform
- Application to Shor's Algorithm
- Live demonstration of Shor's!

This week (`https://arxiv.org/pdf/2408.00075`)
(*Highly-efficient quantum Fourier transformations for certain nonabelian groups*)

- Introduction to group representations
- The nonabelian Fourier transform
- New quantum algorithms

Group theory for chemists

"Groups are symmetries of space."

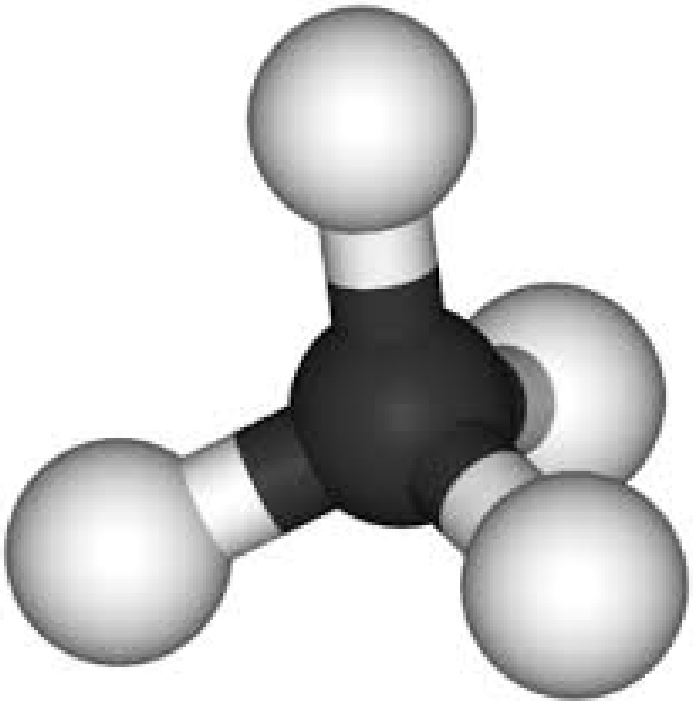


Figure: Tetrahedral symmetry (T)

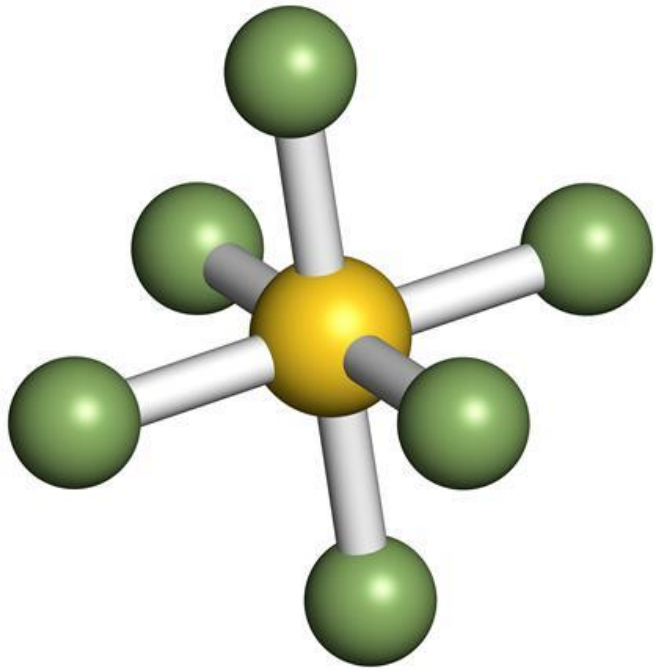


Figure: Octahedral symmetry (O)

The Treachery of Groups



These are not groups, but *representations* of groups.

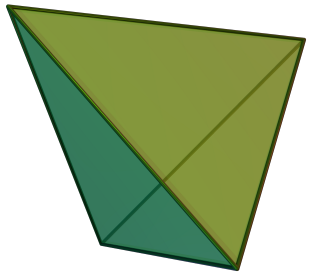
Group definition

Definition

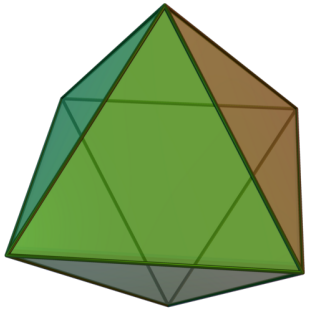
A *group* is a set G equipped with an operation $\times : G \times G \to G$ satisfying

- The existence of 1 and inverses $(g^{-1}) \in G$ such that $g(g^{-1}) = (g^{-1})g = 1$
- Associativity: $a(bc) = (ab)c$

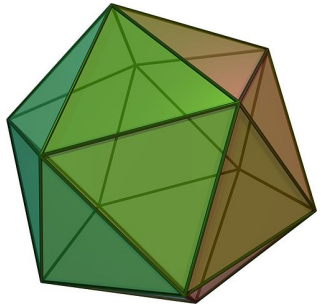
Comparing definitions



A_4: Parity-preserving permutations of 4 points



S_4: General permutations of 4 points



A_5: Parity-preserving permutations of 5 points

Representations

Definition

A *representation* of G is an action of G on a vector space.

More specifically,

Definition

A *representation* is a map $\rho : G \to GL(n)$ (the group of invertible matrices) such that

$$\rho(g_1 g_2) = \rho(g_1)\rho(g_2)$$

$SU(2)$

Recall rotations in quantum mechanics

$$U(\hat{n} \cdot \hat{\sigma})U^{\dagger} = (\rho(U)\hat{n}) \cdot \hat{\sigma}$$

We get a map $SU(2) \to SO(3)$. $SO(3)$ is just a representation of $SU(2)$!

Faithful Representations

$SO(3)$ is not a *faithful representation*.

$$\rho\left(\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\right) = \rho\left(\begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}\right) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Not invertible.

Projective Representations

Definition

A *projective representation* is a representation *up to rescaling*.
Alternatively, a map

$$\rho : G \to PGL(V)$$

$SU(2)$ is a projective representation of $SO(3)$.
In general, this doesn't lift to an ordinary representation.

Example

$$\rho : \mathbb{Z}_2 \to PGL(2, \mathbb{R}) \qquad \rho([1]) = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

Central extensions

Definition

Every projective representation of G lifts to an ordinary representation of a *central extension* of G.

Definition

Given a group G, a *central extension* of G by A is a group E such that

- $E/A \simeq G$
- $A \subseteq Z(E)$ (everything in A commutes with everything in E)

E.g. Virasoro algebra

The Program

Building special subgroups of $SU(2)$:

- Pick a point group $H \in SO(3)$
- Get a projective representation of H in $SU(2)$
- Obtain a central extension of H that's naturally a subgroup of $SU(2)$

Binary Groups

- Binary octahedral $\mathbb{BO}$. Central extension of $O \simeq S_4$
  - $|\mathbb{BO}| = 48$
- Binary tetrahedral $\mathbb{BT}$. Central extension of $T \simeq A_4$
  - $|\mathbb{BT}| = 24$
- Quaternion Q_8. Central extension of $V \simeq \mathbb{Z}_2^2$
  - $|Q_8| = 8$

$$f = (-1)^a \mathbf{j}^b \mathbf{k}^c \tag{17}$$
$$g = (-1)^a \mathbf{j}^b \mathbf{k}^c \mathbf{u}^d \tag{18}$$
$$h = (-1)^a \mathbf{j}^b \mathbf{k}^c \mathbf{u}^d \mathbf{t}^e. \tag{19}$$

The generators which appear in these above equations have the following matrix presentation in the faithful representation:

$$\mathbf{j} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \ \mathbf{k} = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix},$$
$$\mathbf{u} = \begin{pmatrix} -\eta & -\eta \\ \eta^* & -\eta^* \end{pmatrix}, \ \mathbf{t} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -i \\ -i & 1 \end{pmatrix} \tag{20}$$

Why no $\mathbb{BI}$?

solvable			not solvable
D_n	A_4	S_4	A_5

Extremely relevant to pure math.

The Nonabelian Fourier Transform

$$\hat{f}(\rho) = \sum_{g \in G} f(g)\rho(g)$$

Turns a function on G into a function that takes in a representation and outputs a matrix.

- How to compute this?
- What motivates this?

How to compute?

Just as numbers can be factored into primes, we can break any representation of a group G into the sum of *irreducible representations* ("irreps").

Why this definition?

Consider the abelian case (over $\mathbb{C}$)

Magical Fact

Every irrep of an abelian group is 1-dimensional.

Magical Fact

The image of any group element has norm 1.

All representations are elements of $\mathrm{Hom}(G, S^1)$, the Pontryagin dual group!

Implementation

Need to ensure 2 things

- Linear isomorphism
- Unitarity

Linear isomorphism

The Fourier transform is linear.

Magical Fact

$$|G| = \sum_{\rho \text{ irrep}} \dim(\rho)^2$$

$$\begin{array}{cc} |000\rangle & |001\rangle \\ |010\rangle & |011\rangle \\ |100\rangle & |101\rangle \\ |110\rangle & |111\rangle \end{array} \xrightarrow{FT_{Q_8}} \begin{array}{cc} |\xi_1\rangle & |\xi_2\rangle \\ |\xi_3\rangle & |\xi_4\rangle \\ |\xi_5, 1\rangle & |\xi_5, 2\rangle \\ |\xi_5, 3\rangle & |\xi_5, 4\rangle \end{array}$$

Unitarity

Magical Fact

Every representation is isomorphic to a unitary representation

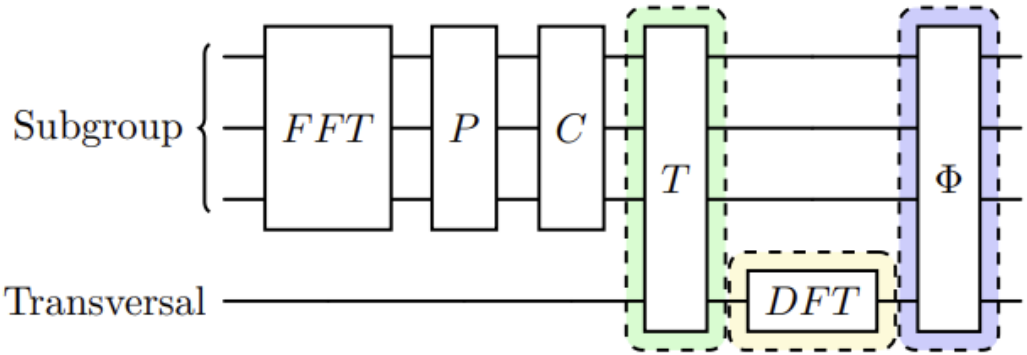
The image of a basis vector $|g\rangle$ should be a direct sum of the image of g under all irreps.

With the correct normalization,

$$U|g\rangle = \frac{1}{\sqrt{|G|}} \sum_{\rho \text{ irrep}} \dim(\rho) |\rho, \rho(g)\rangle$$

Circuit

Relies on subgroup structure



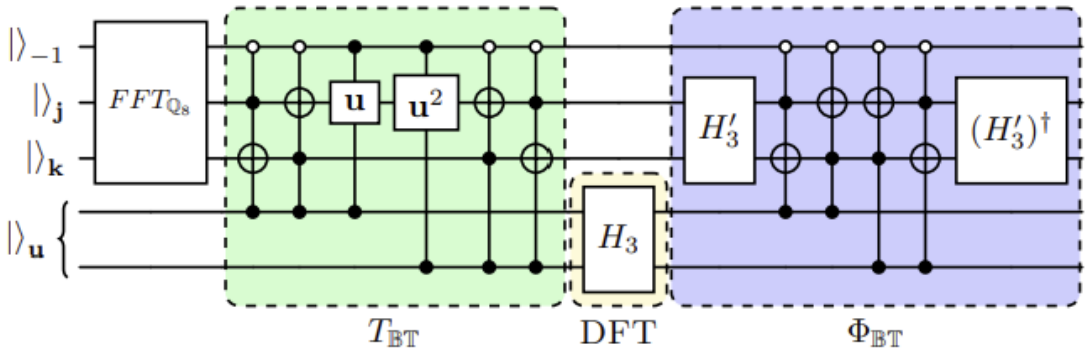


FIG. 6. Quantum circuit for $\mathfrak{U}_{FFT}^{\mathbb{BT}}$. The gate H_3' is a discrete Fourier transformaiton over $\mathbb{Z}_3$ on the $|01\rangle$, $|10\rangle$, $|11\rangle$ register.

$SU(3)$

$$\mathbb{1} \to \mathbb{Z}_3 \times \mathbb{Z}_3 \to \Delta(27) \to \Delta(54) \to \Sigma(36 \times 3)$$
$$\mathbb{1} \to \mathbb{Z}_2 \to \mathbb{Z}_4 \to \mathbb{Q}_8 \to \mathbb{B}\mathbb{T} \to \mathbb{B}\mathbb{O}.$$

The group elements, $g \in \Delta(27)$, $h \in \Delta(54)$, and $f \in \Sigma(3 \times 36)$, can be written as

$$g = \omega_3^p \mathbf{C}^q \mathbf{E}^r \tag{36}$$
$$h = \omega_3^p \mathbf{C}^q \mathbf{E}^r \mathbf{V}^{2s} \tag{37}$$
$$f = \omega_3^p \mathbf{C}^q \mathbf{E}^r \mathbf{V}^{2s+t} \tag{38}$$

Useful in particle physics(?)

Algorithms

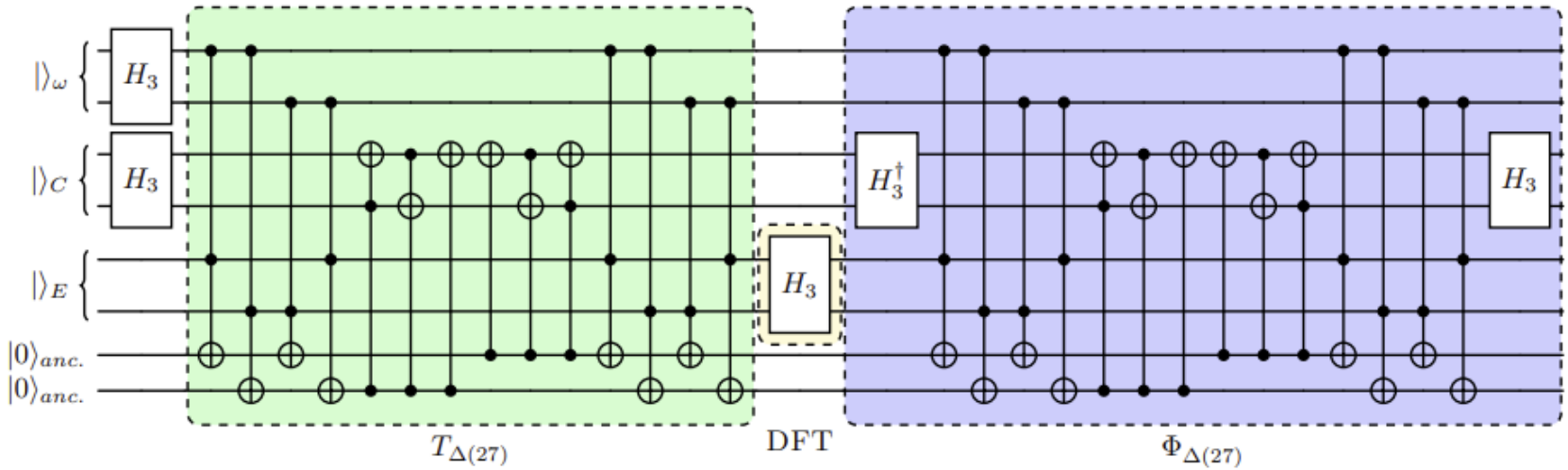
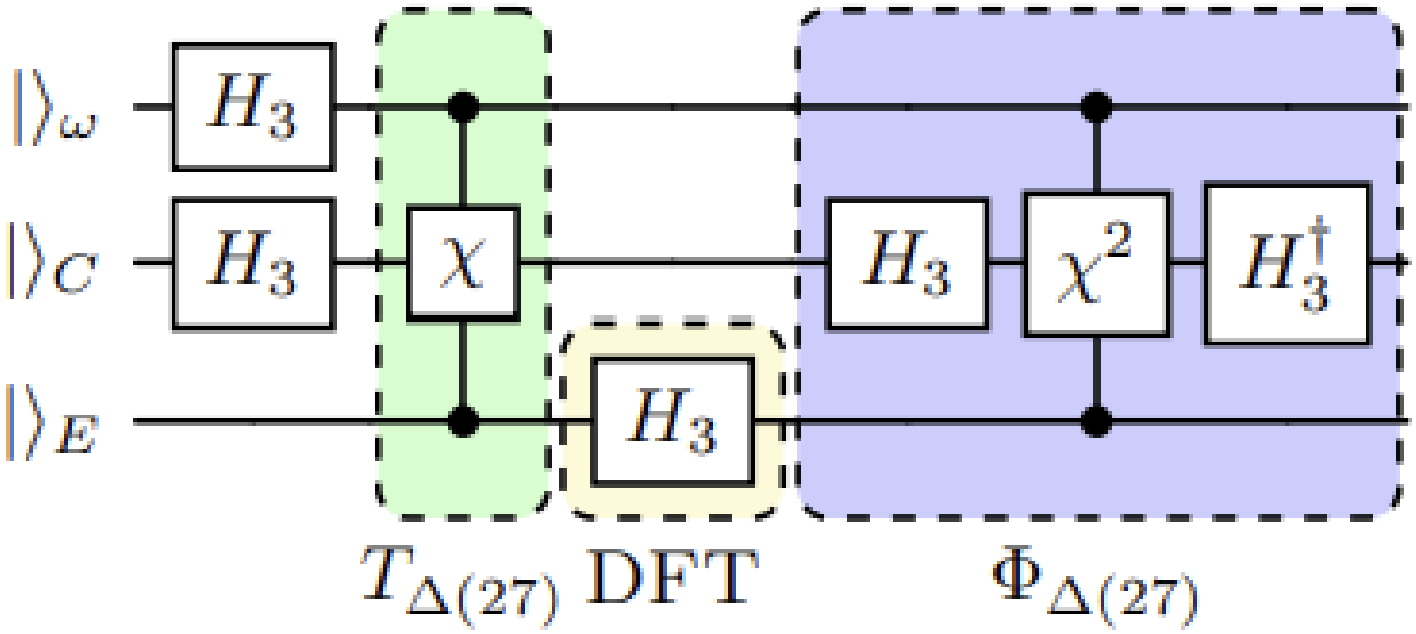


Figure: Algorithm for $\Delta(27)$

Qutrits

Like qubits but 50% better



Mixed architectures

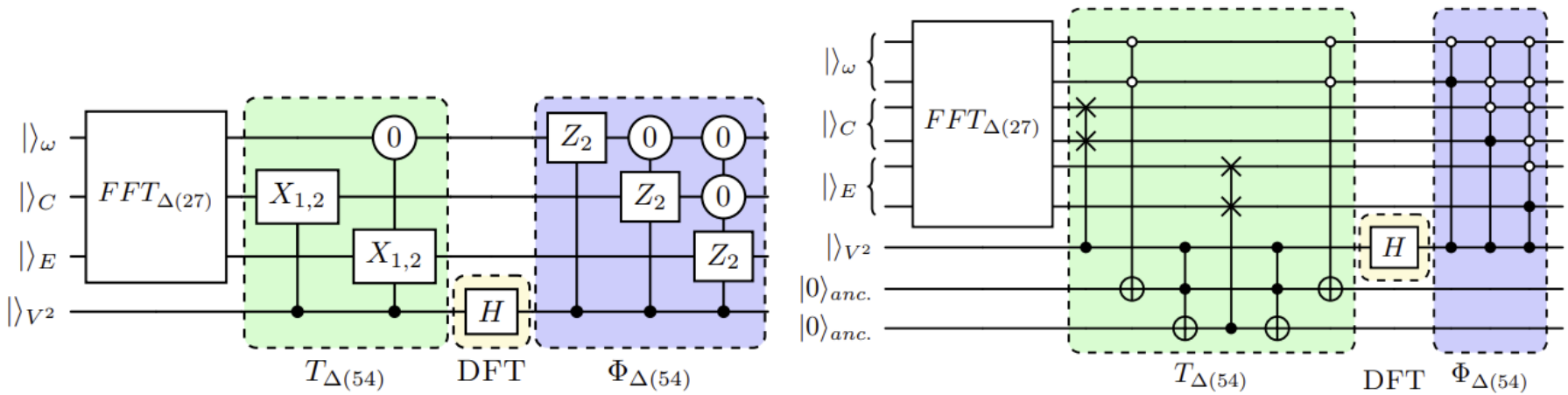


FIG. 9. Quantum circuit of $\mathfrak{U}_{FFT}^{\Delta(54)}$ using a heterogenous qubit-qutrit register (left) and homogenous qubit register (right).

Rest of the paper: similar constructions & mathematical specifics